Response to Reviews of the Paper: Harnessing Frequency-Aware Network for Crack Detection in Road-Perceptive Autonomous Vehicle Scenes

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Dear Associate Editor and Reviewers:

We sincerely thank you for the constructive feedback on our revised manuscript (CYB-E-2025-10-3956). We are encouraged that Reviewer 2 is fully satisfied with the previous revision and recommends acceptance, and we are grateful to the Associate Editor and Reviewer 1 for the remaining suggestions. These suggestions concern visualization representativeness, module presentation, failure analysis, and positioning relative to recent work.

In this second-round revision, we have addressed the remaining comments as detailed below. Specifically, we have (i) clarified the selection criteria of the qualitative visualizations and added exact configuration labels and per-image AIU values to the caption of the frequency-ablation visualization (Fig. 6); (ii) restructured the descriptions of the proposed modules (FAM, T-M, and HFM) in Sections III-B and III-C into an "intuition-first" style, providing a concise high-level explanation of each module before the mathematical formulation and consolidating the previously scattered equation-by-equation commentary; (iii) added a new subsection "Failure Cases and Limitations" (Section IV-H) that presents illustrative failure examples and analyzes the boundary conditions of the proposed method; and (iv) updated the Related Work section with recent 2024–2025 publications from journals and conferences, including papers from IEEE Transactions on Cybernetics.

For clarity of marking, substantive changes that address the remaining comments are highlighted in red in the revised manuscript, while minor grammatical, numerical-presentation, and bibliographic corrections are left unmarked. The red markings of the previous round have been restored to black.

Below, we address each comment from the Associate Editor and the reviewers in turn. The comments are labelled in bold and blue, followed by our responses.

CONTENTS

I. Response to Associate Editor

Comments to the Author:

The revised manuscript has been substantially improved; however, further clarification is still needed regarding the representativeness of the visualizations, presentation of the proposed modules, and discussion of failure cases and limitations. I therefore recommend revision before acceptance. To put your contributions into the context of the most recent research, please update your Related Work section to include relevant references that have appeared in the past two years in conferences and journals, including from this Transaction.

Response: We sincerely appreciate your positive assessment of the previous revision and your time in handling our manuscript. We have addressed the remaining concerns as detailed below:

- **Representativeness of the visualizations.** We now explicitly state the selection criteria of the qualitative figures and, for the frequency-ablation visualization in Fig. 6, label the exact configuration of each panel and report its per-image AIU in the caption. Please see our detailed response to **Reviewer 1, Comment 1**.
- **Presentation of the proposed modules.** Sections III-B and III-C have been restructured in an "intuition-first" style: each module (FAM, T-M, and HFM) now begins with a concise, high-level explanation of its purpose and design rationale before any mathematical expression, and the previously scattered equation-by-equation commentary has been consolidated and condensed. Please see our detailed response to **Reviewer 1, Comment 2**.
- **Failure cases and limitations.** A new subsection, **"Failure Cases and Limitations" (Section IV-H)**, has been added. It presents illustrative failure examples grouped into two observed failure modes, analyzes why FANet struggles in these conditions, and discusses the practical boundary conditions and future directions. Please see our detailed response to **Reviewer 1, Comment 3**.
- **Related Work update.** We have updated the Related Work section with recent publications that appeared in the past two years in journals and conferences, including works from IEEE Transactions on Cybernetics. The newly added text and references are listed below.

In the revised manuscript, the following paragraph has been added to Section II (Related Work):

More recent studies from the past two years further advance both crack detection and frequency-aware representation learning. On the crack-detection side, UP-CrackNet [1] formulates road crack detection as unsupervised adversarial image restoration to remove the dependence on pixel-wise annotations; SCSegamba [2] designs a lightweight structure-aware vision Mamba that captures crack morphology and texture with minimal computational cost; and an ultra-lightweight Mamba network with frequency-domain feature extraction has been developed for pavement tiny-crack segmentation [3]. On the frequency-modeling side, FreqFusion [4] rectifies feature fusion for dense prediction with adaptive low-/high-pass filters, and FSEL [5] entangles frequency and spatial representations for camouflaged object detection. Closely related inspection and segmentation problems have also been actively studied in IEEE Transactions on Cybernetics, including zero-shot industrial anomaly detection with hybrid vision-language prompts [6] and contrastive transformer-based segmentation of subtle structures [7]. These advances confirm the growing interest in spectral cues and efficient architectures; however, they either address generic dense prediction or treat frequency information as an auxiliary enhancement, leaving a deeply coupled frequency-spatial representation dedicated to road crack detection, the very focus of FANet, largely unexplored.

The above paragraph has been added to the end of **Section II (Related Work)**, and the seven newly cited works are: two from IEEE Transactions on Cybernetics (AnomalyVLM, TCYB 2025 [6]; CTNet, TCYB 2024 [7]), three recent crack-detection studies (UP-CrackNet, T-ITS 2024 [1]; SC-Segamba, CVPR 2025 [2]; ultra-lightweight Mamba–frequency network, ESWA 2025 [3]), and two recent frequency-aware representation studies (FreqFusion, TPAMI 2024 [4]; FSEL, ECCV 2024 [5]).

II. Response to Reviewer 1

General Comments:

In this paper, the authors propose FANet, a frequency–spatial synergistic framework for crack detection that integrates a Frequency-Aware Module (FAM), a Tri-Merge decoder (T-M), and a Hybrid Fusion Module (HFM). The revised manuscript provides expanded technical explanations, ablation studies involving low-/high-/full-frequency variants of FAM, and frequency-based visual comparisons. The work addresses a meaningful problem in pavement crack analysis and shows clear improvements over baseline models. However, several issues still require clarification or refinement before the paper can be accepted.

Response: We sincerely thank the reviewer for the careful reading of our revised manuscript, the accurate summary of our contributions, and the constructive suggestions. We address the three remaining comments as detailed below.

Q1 Comments to the Author:

The manuscript includes frequency-based visualizations (high/low/fusion), but the representativeness and selection criteria of these examples should be clarified. The authors should explicitly state whether the provided samples are typical, randomly selected, or chosen as illustrative cases, and each visualization should clearly label the specific configuration used. Adding the corresponding quantitative metric of each visualized sample to the caption would help readers connect qualitative and quantitative evidence more directly.

Response: We thank the reviewer for this valuable suggestion. In the revised manuscript, we have made three changes to clarify the representativeness of the frequency-based visualizations:

1) **Selection criteria stated explicitly.** We now explicitly state in Section IV-G that the sample shown in the frequency-based visualization (Fig. 6) is a *purposefully selected illustrative case* (not randomly sampled, and not a best-performing case): it simultaneously contains a wide trunk crack and hairline branches on coarsely textured pavement, which is the condition under which the differences among the three FAM configurations are most visible. A corresponding statement of the selection criteria has also been added for the qualitative comparison figure (Fig. 5) in Section IV-F.
2) **Configuration of each panel labelled.** The caption of Fig. 6 now explicitly maps each panel to the exact configuration used in the ablation study (Table II): panel (c) is produced by *Baseline + FAM (Full)*, panel (d) by *Baseline + FAM (High-Freq)*, and panel (e) by *Baseline + FAM (Low-Freq)*.
3) **Per-image metrics added to the caption.** The per-image AIU of each visualized prediction is now reported in the caption of Fig. 6, computed with the same evaluation code as Table II, complementing the dataset-level evidence in Table II.

In the revised manuscript, the following text has been added to Section IV-G (Ablation Studies):

To clarify the representativeness of the frequency-based visualization, the sample in Fig. 6 is a purposefully selected illustrative case from the Crack500 test set, rather than a random or best-performing sample. It contains a wide trunk crack and hairline branches on coarsely textured pavement, where the differences among the three FAM configurations are readily visible. Panels (c)–(e) correspond to *Baseline + FAM (Full)*, *Baseline + FAM (High-Freq)*, and *Baseline + FAM (Low-Freq)* in Table II, respectively; the caption reports a per-image AIU for each panel, computed with the same evaluation code as Table II over the threshold set $T = \{0.01, 0.02, \ldots, 0.99\}$.

The caption of Fig. 6 has been rewritten as follows (the figure is reproduced below; figure/table numbers in the quoted text refer to the revised manuscript):

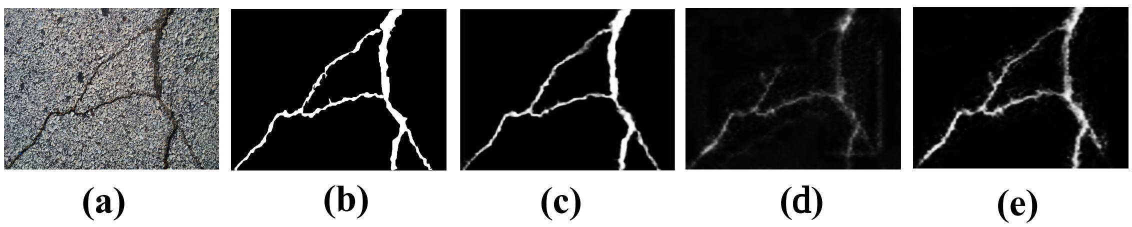

Fig. 6. Qualitative comparison of three FAM configurations in Table II on a purposefully selected illustrative case from the Crack500 test set. (a) Input. (b) Ground truth. (c) Prediction of *Baseline + FAM (Full)*, which fuses the high- and low-frequency branches; per-image AIU = 0.62. (d) Prediction of *Baseline + FAM (High-Freq)*; per-image AIU = 0.45. (e) Prediction of *Baseline + FAM (Low-Freq)*; per-image AIU = 0.41. This illustrative example qualitatively shows the complementary behavior of the two frequency branches; dataset-level conclusions are based on Table II.

In addition, the following sentence has been added to Section IV-F (Qualitative Results) to state the selection criteria of the method-comparison figure (Fig. 5):

The examples in Fig. 5 are purposefully selected illustrative cases from the five test sets. They cover thin hairline cracks, coarse pavement texture, shadows and uneven illumination, and crack-like distractors; they are not random or best-performing samples, and all methods are compared on identical inputs.

Q2 Comments to the Author:

The overall descriptions of FAM, T-M, and HFM—particularly in Sections III-B and III-C—remain verbose and would benefit from a clearer separation between intuition and formalism. Providing a concise, high-level explanation of each module's purpose before introducing the mathematical expressions would significantly improve readability and help interdisciplinary readers better understand the design motivations.

Response: We thank the reviewer for this insightful suggestion, which has significantly improved the readability of the method section. In the revised manuscript, Sections III-B and III-C have been restructured following exactly the principle suggested by the reviewer, namely a clear separation between intuition and formalism:

1) **Intuition first.** Each of the three modules (FAM, T-M, and HFM) now begins with a concise "*Design intuition*" paragraph that explains, in plain language and without any equation, what problem the module solves and why it is designed this way.

2) **Formalism afterwards, condensed.** The mathematical formulation follows the intuition paragraph. The previously scattered equation-by-equation commentary (several separate explanatory passages inserted after Eqs. (1)–(4) in the last revision) has been consolidated into brief remarks, and overlapping explanations of the FAM mechanism, which appeared three times in slightly different wording, have been merged into a single passage. As a result, Sections III-B and III-C are now *shorter* than in the previous version while conveying the same technical content.

In the revised manuscript, the newly added "Design intuition" paragraphs are shown below:

(Section III-B, before the formulation of FAM) Design intuition of FAM. Road cracks appear as sparse and sharp intensity discontinuities embedded in slowly varying pavement context. In the feature space, the former concentrates in the high-frequency band, whereas the latter dominates the low-frequency band. The FAM is therefore designed to (i) decompose each backbone feature into a high-frequency and a low-frequency component, (ii) let the two components exchange complementary information, so that edge responses acquire contextual support and contour responses are cleansed of texture noise, and (iii) re-weight and merge the two bands using learnable weights. The output is a frequency-balanced feature in which crack evidence is amplified relative to background texture. The remainder of this subsection formalizes these three steps.

(Section III-B, before the formulation of the T-M decoder) Design intuition of T-M. The T-M decoder aggregates the frequency-aware features of the three deepest levels: deeper features indicate *where* a crack is likely to exist, while shallower ones describe its shape more precisely. Multiplying adjacent feature maps promotes responses that are jointly activated across semantic levels, thereby suppressing level-specific noise and producing a cleaner coarse localization map S_1.

(Section III-C, before the formulation of HFM) Design intuition of HFM. The HFM addresses two questions during refinement. The first is *where to refine*: at refinement stage i, the current guidance map S_i acts as a spatial prior that amplifies crack-relevant regions of the deeper feature. The second is *what to keep*: a cross-layer feature interaction between adjacent layers determines which high-resolution details are consistent with this prior and should be preserved. These operations refine accuracy with a moderate throughput cost, as quantified in Table II. The following formulation details the two steps.

In addition, the redundant explanatory passages of the previous round have been merged and condensed (the removed redundancies are not marked in red, to keep the marking readable). We believe the method section is now significantly easier to follow, especially for interdisciplinary readers.

Q3 Comments to the Author:

The paper would benefit from a short subsection discussing typical failure cases or limitations. Presenting a small number of representative failure examples and briefly analyzing why FANet struggles in those conditions would provide a more balanced understanding of the model's behavior and practical boundary conditions.

Response: We thank the reviewer for this constructive suggestion, which makes the experimental analysis more balanced. In the revised manuscript, we have added a new subsection, **"Failure Cases and Limitations" (Section IV-H)**, placed after the ablation studies. It presents selected failure examples from FANet's own predictions, grouped into two observed failure modes, analyzes the underlying reasons within the frequency-aware design, and summarizes the practical boundary conditions and future directions of FANet.

In the revised manuscript, the newly added subsection is shown below (figure numbers in the quoted text refer to the revised manuscript, where the failure-case figure is Fig. 7; it is reproduced below):

IV-H. Failure Cases and Limitations. Although FANet achieves the highest values among the methods compared in Table I, it still struggles under two identifiable conditions. Fig. 7 shows two illustrative failure cases selected from the test sets. In each column, the top row is the input image, the middle row is the ground truth, and the bottom row is our prediction, so that the gap between the ground truth and our result can be read directly.

(1) Fragmentation of low-contrast cracks. When the local contrast between a crack and the surrounding pavement is weak, some crack segments receive low prediction confidence. As shown in Fig. 7(a), the ground-truth crack forms a continuous structure, whereas the prediction exhibits weak and uneven responses along the crack, so that several portions are barely activated and the predicted crack loses continuity.

(2) Under-segmentation of fine hairline cracks. When cracks are only a few pixels wide, their responses can be weak relative to the surrounding pavement texture. As shown in Fig. 7(b), the main crack is recovered, but several fine branches visible in the ground truth are missed or only partially detected.

These observations suggest that the current design is least reliable when crack evidence is not a sufficiently distinct sparse high-frequency signal, particularly at very low contrast or very small width. In addition, two limitations are worth noting. First, the two-step coarse-to-fine pipeline reduces throughput by 22.7% (from 13.2 FPS for the PVT baseline to 10.2 FPS for the complete model) on the same hardware, as reported in Table II. Second, FANet is trained with binary cross-entropy only and imposes no explicit topological continuity constraint, which may contribute to the fragmented predictions above. We plan to explore topology-aware objectives, contrast-adaptive frequency weighting, and complementary sensing modalities such as depth or infrared imaging to push these boundaries in future work.

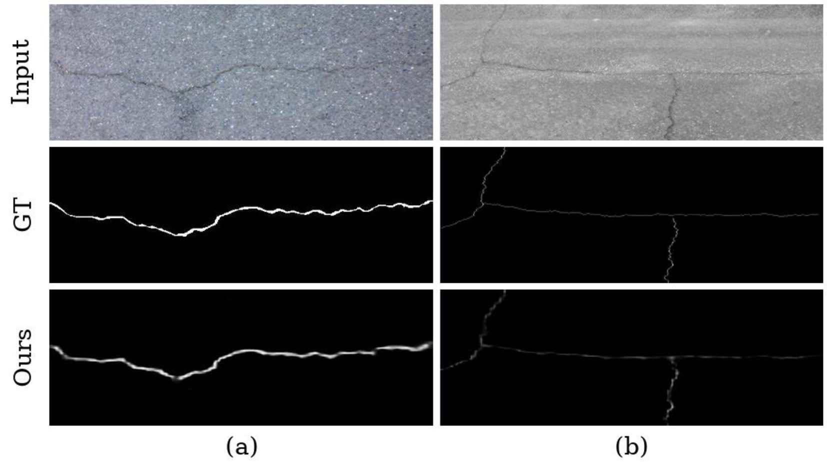

Fig. 7. Illustrative failure cases of FANet. In each column, the top, middle, and bottom rows show the input image, the ground truth, and our prediction, respectively. (a) A low-contrast crack whose continuous ground-truth structure is only weakly and unevenly recovered, losing continuity at several locations. (b) A fine hairline crack whose main structure is recovered while several fine branches are missed or only partially detected.

III. Response to Reviewer 2

Comments to the Author:

The authors have addressed all my concerns. I recommend acceptance. Thanks.

Response: We sincerely thank the reviewer for the positive evaluation of our revised manuscript and for recommending acceptance. The reviewer's constructive comments in the previous round have contributed greatly to improving the quality of this work.

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